Bags, Vectors & Transformers

Working with Other Languages · Bonus, Part 17

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Three Routes to Other Languages

Route	Works well when	Watch out for
A model for your language RobBERT, BERTje for Dutch	one exists, trained on lots of text	only a few dozen languages have one
A multilingual model XLM-R, mDeBERTa, mmBERT	your languages were in its training data	shared capacity: each language gets less
Translate first NLLB-200, DeepL	translation into English is good	lost nuance; commercial services see your texts

LLMs: the same caveat, stronger. They sound fluent in languages they handle badly.

Any route: **validate in every language**, with native-speaker coders.

One Set of Sentences, Nine Languages

SIB-200 (Adelani et al., 2024): the same 1,004 sentences from Wikinews, Wikijunior and Wikivoyage, professionally translated into 200+ languages, each labeled with one of seven topics.

Because the data is **parallel**, every comparison is fair: the content is identical, only the language changes. Any difference between languages is the model's doing.

Well served: English, Dutch, German

Small regional languages: Limburgish, Luxembourgish

Millions of speakers, little digital text: Swahili, Yoruba, Amharic (Ge'ez script), Ayacucho Quechua

Tokenizers: Same Content, Different Costs

A tokenizer built mostly from English cuts other languages into many more, smaller pieces. The same content can take **up to 15 times as many tokens** (Petrov et al., 2023): more cost with a paid API, more time, less text in the context window.

In a script it barely knows, an English tokenizer falls back to single characters or to `[UNK]` (unknown): then the text is simply gone.

In the notebook: tokens needed for the same 204 sentences, relative to English, for five tokenizers (English, Dutch, XLM-R, mmBERT, an LLM) in nine languages.

Quick check for your own data: tokenize a few sentences and look at the pieces. If words shatter into letters, the model knows the language poorly.

Dutch, and Train in English, Apply Everywhere

Dutch: RobBERT vs. XLM-R vs. TF-IDF

A good monolingual model is usually at least as good as a multilingual one for its own language: the "curse of multilinguality" (Conneau et al., 2020).

Zero-shot cross-lingual transfer

Fine-tune XLM-R on English labels, apply it to eight other languages. Expect a staircase, lowest for languages barely seen in pretraining.

In the SIB-200 study, transfer from English stayed below 70% accuracy for 81 of 205 languages, worst for languages from Africa, the Americas and Oceania: the speakers already least served are measured worst. Transfer works best **within a language family and script** (Dutch for Limburgish).

Translate First, or Translate the Training Data

NLLB-200 (NLLB Team, 2024): an open translation model for 200 languages that runs on Colab, so texts never leave your machine. Research use only.

- **translate-test:** your texts into English, then an English model
- **translate-train:** English training data into your language, then fine-tune

For bag-of-words methods, machine translation works well for comparative research in major European languages (de Vries et al., 2018).

Measure the translation, too. With parallel data you know the correct English, so chrF (character overlap, 0–100) scores every translation.

Translation helps where the classifier reads the language badly **and** the translator reads it well. Where translation is poor, it only moves the problem.

Low-Resource Languages: What Helps

1. **Check** the model's languages and its tokenizer; try regional models (AfroXLMR, IndicBERT)
2. **Continue pretraining** on your unlabeled texts (Gururangan et al., 2020)
3. **Transfer from a related language**, not from English
4. **Translate** data, and have samples checked
5. **Be extra careful with LLMs**: weaker, still fluent
6. **Work with native speakers** throughout

"Low-resource" means little digital text, few labeled datasets, weak tools. It covers most of the world's languages, including many with tens of millions of speakers.

Dialects, non-standard spelling and languages written in another script (Arabic in Latin letters) behave like low-resource languages even inside big ones.

Corpora with Many Languages

Which language is it?
Lingua (75 languages) and GlotLID (2,000+, low-resource). A tool only knows its own languages; the rest get filed under a neighbor, silently.

Same meaning everywhere?
On parallel data the true topic shares are identical, so differences in predicted shares are pure measurement error.

Meaning or language?
Embeddings often group texts by language first. Is a sentence's nearest neighbor its translation?

For comparative research: validate in every language, report accuracy per language, and only interpret differences larger than the differences in accuracy.

In the Notebook

- SIB-200: the same sentences in nine languages, from English to Quechua
- Five tokenizers, and what the same content costs in each language
- Dutch: RobBERT, XLM-R and a TF-IDF baseline
- Cross-lingual transfer, translate-test and translate-train with NLLB-200
- Language identification with Lingua and GlotLID, also on short texts
- Is the measure comparable across languages? Do embeddings group by meaning?

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